

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — MASTER TOPICAL PRACTICE PACK: 11A

FURTHER KINETICS (RATES, ORDERS, MECHANISMS & ARRHENIUS)

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 11: KINETICS 2
SUB-TOPIC:	11A: Further Kinetics (Rates, Orders, Mechanisms & Arrhenius)
TOTAL QUESTIONS:	50 Authentic Past Paper Questions (Tier 1 & Tier 2)
TOTAL MARKS:	213 Marks
EXAMINER FAQs:	10 Empirical Examiner Traps & Model Answers Included

HOW TO USE THIS A* TOPICAL PACK:

1. Tier 1 (Q1–Q25): Mid-Easy to Moderate foundational questions (1–3 marks, MCQs, core calculations).
2. Tier 2 (Q26–Q50): Hard & A* Challenge questions (4–6 marks, multi-step calculations, multi-spectral NMR, extended answers).
3. Review the complete worked mark scheme in Section C after attempting each tier.
4. Master the 10 Empirical Examiner FAQs in Section D to eliminate common candidate traps.

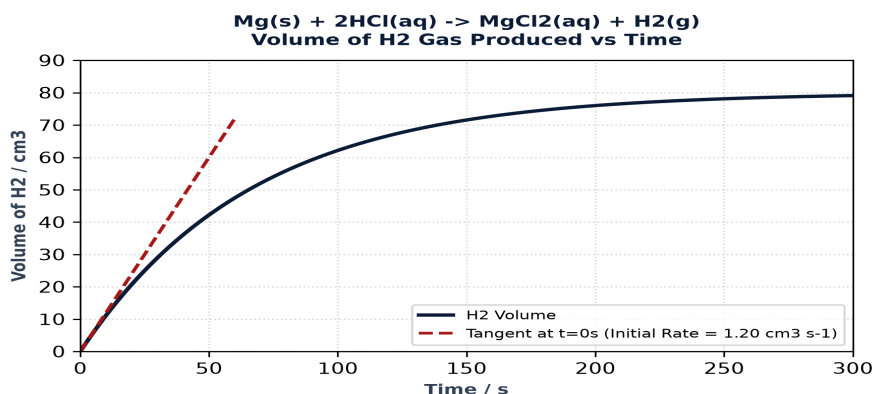
SECTION A: TIER 1 PRACTICE QUESTIONS (FOUNDATIONAL TO MODERATE) — Q1 to Q25

Q1. 1. Continuous Monitoring of Reaction Rate

[4 Marks]

WCH14/01/Jan23/Q1

The reaction between magnesium ribbon and dilute hydrochloric acid was investigated:



(a) State the most suitable experimental method to continuously monitor the rate of this reaction and explain your choice. [2]

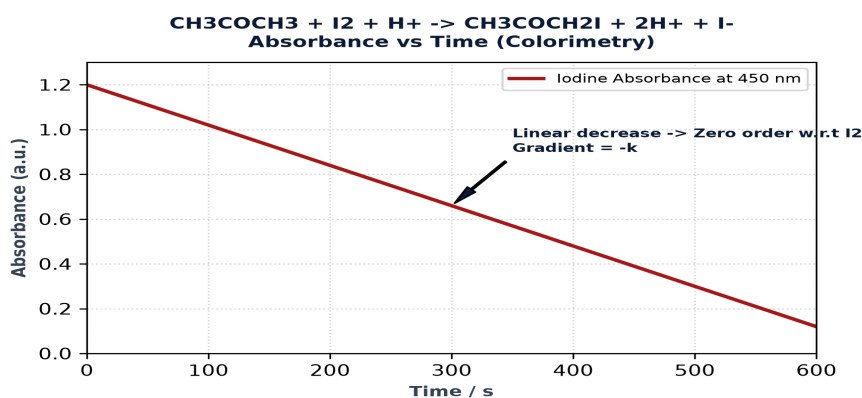
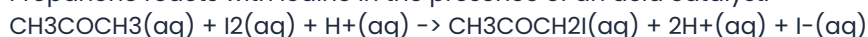
(b) Explain how the initial rate of reaction is determined from the gas volume vs time graph. [2]

Q2. 2. Colorimetric Measurement of Reaction Rate

[3 Marks]

WCH14/01/Oct22/Q2

Propanone reacts with iodine in the presence of an acid catalyst:



(a) Explain how colorimetry can be used to follow the progress of this reaction. [2]

(b) State why colorimetry is preferred over titrametric quenching in this experiment. [1]

Q3. 3. Quenching and Titrimetry

[4 Marks]
WCH14/01/Jun22/Q3

The alkaline hydrolysis of ethyl ethanoate is represented by:
 $\text{CH}_3\text{COOCH}_2\text{CH}_3(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{CH}_3\text{COO}^-(\text{aq}) + \text{CH}_3\text{CH}_2\text{OH}(\text{aq})$

(a) Explain why quenching is necessary before titrating samples taken from this reaction mixture. [2]

(b) Suggest a suitable quenching method for this reaction. [2]

Q4. 4. Electrical Conductivity Method

[3 Marks]
WCH14/01/Jan22/Q4

The bromination of propanone produces H^+ and Br^- ions:
 $\text{CH}_3\text{COCH}_3 + \text{Br}_2 + \text{H}^+ \rightarrow \text{CH}_3\text{COCH}_2\text{Br} + 2\text{H}^+ + \text{Br}^-$

(a) Explain why electrical conductivity increases as the reaction proceeds. [2]

(b) State how conductivity data is converted into rate data. [1]

Q5. 5. Rate Equation and Rate Constant Units

[3 Marks]
WCH14/01/Oct21/Q1

A reaction $\text{A} + 2\text{B} \rightarrow \text{C}$ has the rate equation: $\text{rate} = k[\text{A}][\text{B}]^2$

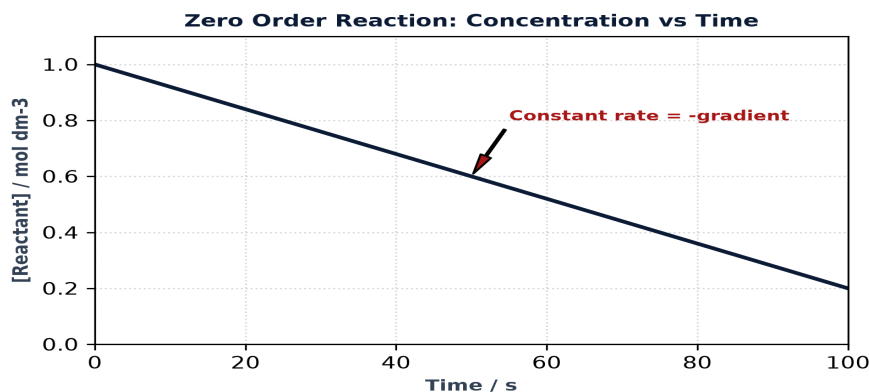
(a) State the order of reaction with respect to A, B, and the overall order. [2]

(b) Deduce the units of the rate constant, k. [1]

Q6. 6. Zero Order Kinetics Definition

[2 Marks]
WCH14/01/Jun21/Q2

The rate equation for a reaction $\text{X} \rightarrow \text{Y}$ is $\text{rate} = k[\text{X}]^0$



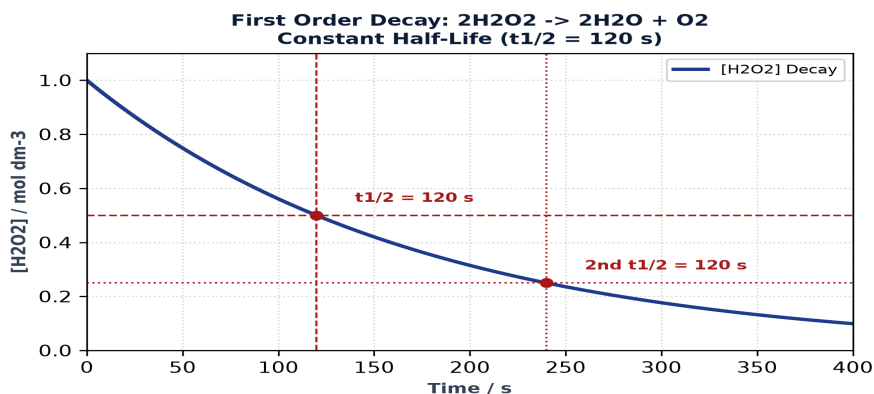
(a) Explain what zero order with respect to reactant X implies about the reaction mechanism. [1]

(b) Describe the shape of the concentration-time graph for a zero-order reaction. [1]

Q7. 7. First Order Half-Life Relationship

[3 Marks]
WCH14/01/Jan21/Q3

The decomposition of hydrogen peroxide $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$ is first order.



(a) Define half-life ($t_{1/2}$) for a chemical reaction. [1]

(b) Explain how the concentration-time graph confirms that a reaction is first order. [2]

Q8. 8. Calculation of Rate Constant from Half-Life

[3 Marks]
WCH14/01/Oct20/Q4

For a first-order decomposition reaction, the half-life $t_{1/2}$ is 450 seconds.

(a) Calculate the rate constant, k , using $t_{1/2} = \ln 2 / k$. Include units. [2]

(b) Calculate the rate when $[\text{reactant}] = 0.080 \text{ mol dm}^{-3}$. [1]

Q9. 9. Initial Rates Method: Deduction of Orders

[4 Marks]

WCH14/01/Jun20/Q5

Initial rate data for $2\text{NO}(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{NO}_2(\text{g})$:

Exp 1: $[\text{NO}] = 0.010$, $[\text{O}_2] = 0.010$, $\text{Rate} = 1.4 \times 10^{-4}$

Exp 2: $[\text{NO}] = 0.020$, $[\text{O}_2] = 0.010$, $\text{Rate} = 5.6 \times 10^{-4}$

Exp 3: $[\text{NO}] = 0.010$, $[\text{O}_2] = 0.030$, $\text{Rate} = 4.2 \times 10^{-4}$

(a) Deduce the order w.r.t NO and O₂ with full reasoning. [3]

(b) Write the overall rate equation. [1]

Q10. 10. Calculation of Rate Constant k from Experimental Table

[3 Marks]

WCH14/01/Jan20/Q6

Using data from Q9 (Exp 1: $[\text{NO}] = 0.010$, $[\text{O}_2] = 0.010$, $\text{Rate} = 1.4 \times 10^{-4} \text{ mol dm}^{-3} \text{ s}^{-1}$):

(a) Calculate the value of the rate constant, k. [2]

(b) State the units of k. [1]

Q11. 11. Rate-Determining Step Deduction from Rate Equation

[4 Marks]

WCH14/01/Oct19/Q7

The reaction $\text{NO}_2(\text{g}) + \text{CO}(\text{g}) \rightarrow \text{NO}(\text{g}) + \text{CO}_2(\text{g})$ has rate = $k[\text{NO}_2]^2$.

(a) Explain why CO does not appear in the rate equation. [2]

(b) Propose a two-step mechanism consistent with this rate equation. [2]

Q12. 12. Catalysis & Rate Equation Inclusion

[3 Marks]

WCH14/01/Jun19/Q8

In the acid-catalysed reaction of propanone with iodine, rate = $k[\text{CH}_3\text{COCH}_3][\text{H}^+]$.

(a) Explain why H⁺ appears in the rate equation despite being a catalyst. [2]

(b) State the order with respect to iodine. [1]

Q13. 13. Initial Rates Table with Fractional / Higher Orders

[4 Marks]
WCH14/01/Jan19/Q9

Reaction: $A + B \rightarrow C$

Exp 1: $[A]=0.20$, $[B]=0.20$, Rate= 2.0×10^{-3}

Exp 2: $[A]=0.40$, $[B]=0.20$, Rate= 8.0×10^{-3}

Exp 3: $[A]=0.20$, $[B]=0.40$, Rate= 2.0×10^{-3}

(a) Deduce the order with respect to A and B. [3]

(b) Calculate k including units. [1]

Q14. 14. Multi-Step Mechanism Validation

[4 Marks]
WCH14/01/Sample/Q14

Consider the reaction: $2H_2 + 2NO \rightarrow N_2 + 2H_2O$

Proposed mechanism:

Step 1: $2NO \rightarrow N_2O_2$ (fast)

Step 2: $N_2O_2 + H_2 \rightarrow N_2O + H_2O$ (slow)

Step 3: $N_2O + H_2 \rightarrow N_2 + H_2O$ (fast)

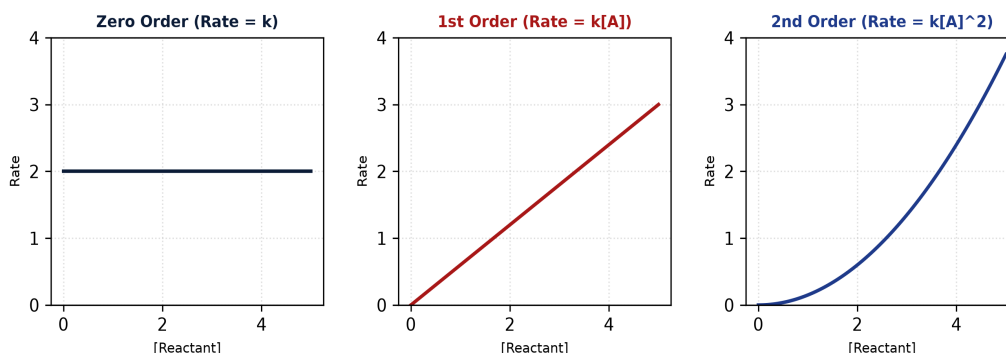
(a) Deduce the rate equation predicted by this mechanism. [3]

(b) Identify any intermediate species. [1]

Q15. 15. Rate-Concentration Graph Summary

[3 Marks]
WCH14/01/Sample/Q15

Distinguish between Zero, 1st, and 2nd order Rate vs Concentration graphs.



(a) Sketch and describe the Rate vs $[A]$ graphs for 0, 1st, and 2nd order. [3]

[4 Marks]

Q16. 16. Hydrolysis of Tertiary Halogenoalkanes Mechanism

WCH14/01/sample/Q16

The hydrolysis of $(\text{CH}_3)_3\text{CBr}$ with NaOH is first order overall: $\text{rate} = k[(\text{CH}_3)_3\text{CBr}]$.

(a) Explain why OH^- does not appear in the rate equation ($\text{S}_\text{N}1$ mechanism). [2]

(b) Write the equation for the rate-determining step. [2]

[4 Marks]

Q17. 17. Hydrolysis of Primary Halogenoalkanes Mechanism

WCH14/01/sample/Q17

The hydrolysis of $\text{CH}_3\text{CH}_2\text{Br}$ with NaOH is second order overall: $\text{rate} = k[\text{CH}_3\text{CH}_2\text{Br}][\text{OH}^-]$.

(a) Explain why this reaction follows an $\text{S}_\text{N}2$ mechanism. [2]

(b) Describe the transition state formed during the rate-determining step. [2]

[3 Marks]

Q18. 18. Effect of Temperature on Rate Constant k

WCH14/01/sample/Q18

As temperature increases from 298 K to 308 K, the rate of a reaction doubles.

(a) Explain why a small increase in temperature causes a large increase in rate. [2]

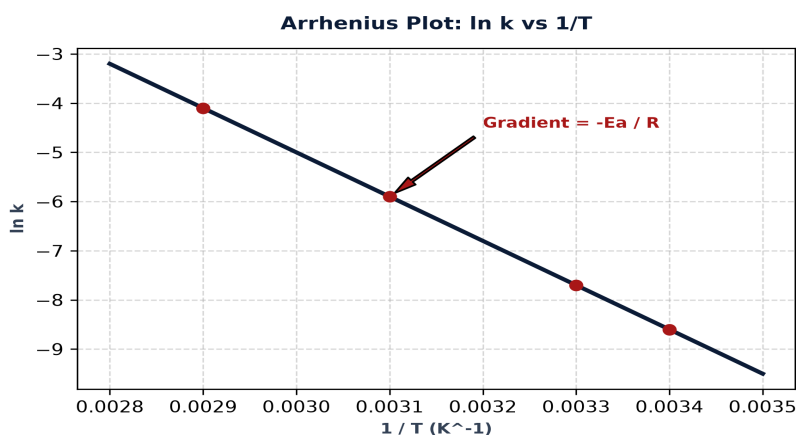
(b) State the relationship between temperature and the rate constant k . [1]

Q19. 19. Arrhenius Equation Formulation

[3 Marks]

WCH14/01/sample/Q19

The Arrhenius equation is $k = A e^{(-E_a / RT)}$.



(a) State the meaning of terms A , E_a , R , and T . [2]

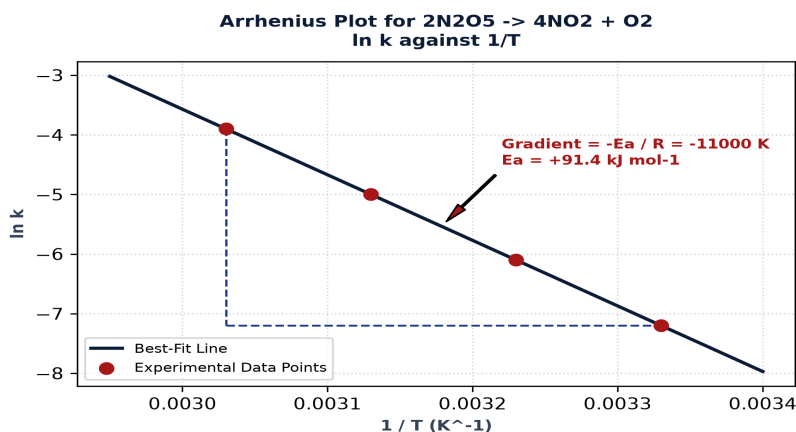
(b) Write the logarithmic form of the Arrhenius equation ($\ln k$ vs $1/T$). [1]

Q20. 20. Arrhenius Plot Graphical Analysis: N_2O_5 Decomposition

[5 Marks]

WCH14/01/Oct19/Q8

Data for the decomposition of N_2O_5 yielded an Arrhenius plot of $\ln k$ against $1/T$ with a straight line gradient of -11000 K .



(a) Calculate the activation energy, E_a , in kJ mol^{-1} . ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$) [3]

(b) If the y-intercept is +29.4, calculate the pre-exponential factor A. [2]

Q21. 21. Heterogeneous Catalysis Surface Adsorption

[3 Marks]

WCH14/01/Sample/Q21

In the Haber process: $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$ using an iron catalyst.

(a) Describe the steps involved in heterogeneous catalysis on the iron surface. [3]

Q22. 22. Homogeneous Catalysis Mechanism: Fe^{2+} and $\text{S}_2\text{O}_8^{2-}$

[4 Marks]

WCH14/01/Sample/Q22

The reaction $\text{S}_2\text{O}_8^{2-} + 2\text{I}^- \rightarrow 2\text{SO}_4^{2-} + \text{I}_2$ is catalysed by $\text{Fe}^{2+}(\text{aq})$ ions.

(a) Explain why the uncatalysed reaction between $\text{S}_2\text{O}_8^{2-}$ and I^- has a high activation energy. [1]

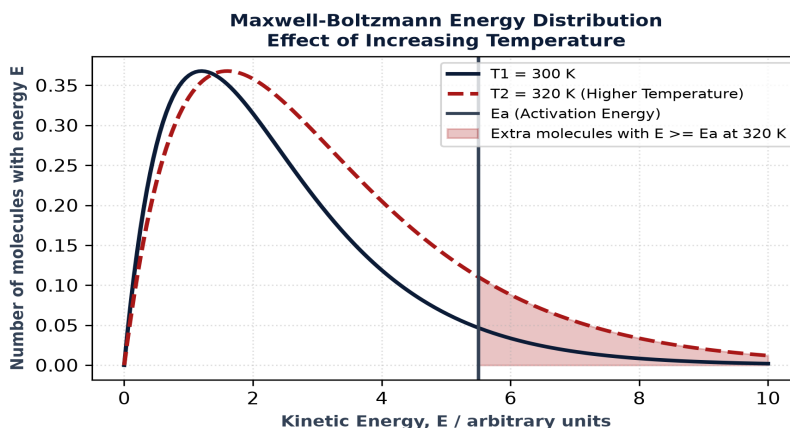
(b) Write two ionic equations showing how Fe^{2+} acts as a catalyst. [3]

Q23. 23. Maxwell-Boltzmann Temperature Shift Analysis

[4 Marks]

WCH14/01/Sample/Q23

The diagram shows Maxwell-Boltzmann distributions at T_1 (300 K) and T_2 (320 K).



(a) Explain why the peak shifts to the right and lowers at higher temperature. [2]

(b) Shade and explain the area representing molecules with E $\geq$ Ea at T2. [2]

Q24. 24. Autocatalysis Mechanism: MnO_4^- and $\text{C}_2\text{O}_4^{2-}$

[4 Marks]

WCH14/01/Sample/Q24

The reaction between acidified KMnO_4 and ethanedioic acid ($\text{H}_2\text{C}_2\text{O}_4$) is autocatalysed by Mn^{2+} .

(a) Define autocatalysis. [1]

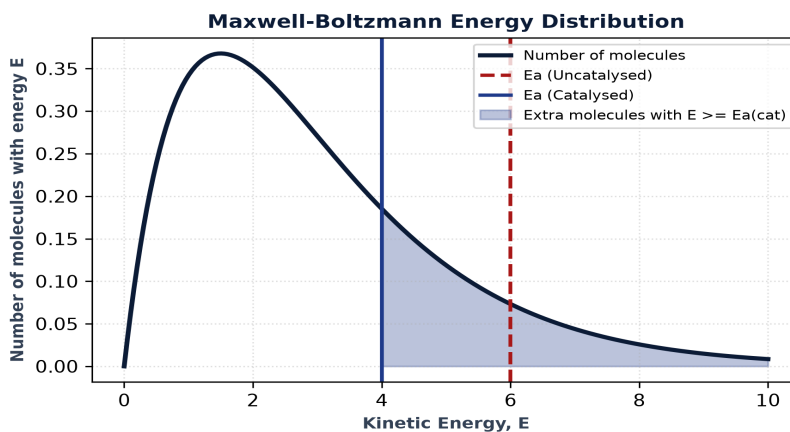
(b) Describe the rate of reaction over time (sigmoidal rate profile). [3]

Q25. 25. Maxwell-Boltzmann Catalysed Activation Energy Barrier

[3 Marks]

WCH14/01/Sample/Q25

A catalyst lowers activation energy from $E_a(\text{uncat})$ to $E_a(\text{cat})$.

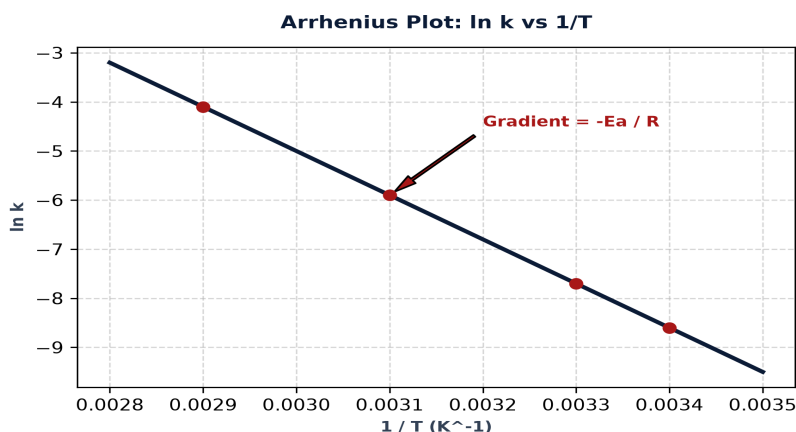


(a) Explain using a Maxwell-Boltzmann curve how a catalyst increases reaction rate without changing temperature. [3]

SECTION B: TIER 2 A* CHALLENGE QUESTIONS (HARD & EXTENDED) — Q26 to Q50

Q26. 26. A* Challenge: Two-Point Arrhenius Activation Energy Calculation**[5 Marks]**
WCH14/01/Hard/Q26

Rate constants for a reaction were measured at two temperatures:

At $T_1 = 300 \text{ K}$, $k_1 = 2.50 \times 10^{-4} \text{ s}^{-1}$.At $T_2 = 340 \text{ K}$, $k_2 = 1.25 \times 10^{-2} \text{ s}^{-1}$. $(R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1})$ (a) Use $\ln(k_2/k_1) = (E_a/R)(1/T_1 - 1/T_2)$ to calculate the activation energy E_a in kJ mol^{-1} . [4](b) Calculate the pre-exponential factor A at 300 K . [1]**Q27. 27. A* Challenge: Deduction of Complex Rate Equation from Mechanism****[5 Marks]**
WCH14/01/Hard/Q27Reaction: $2A + B \rightarrow C + D$

Proposed Mechanism:

Step 1: $A + A \rightleftharpoons A_2$ (fast equilibrium, K_1)Step 2: $A_2 + B \rightarrow C + E$ (slow, rate constant k_2)Step 3: $E \rightarrow D$ (fast)(a) Deduce the overall rate equation in terms of $[A]$ and $[B]$. [3](b) State the overall order and units of the rate constant k . [2]

Q28. 28. A* Challenge: Initial Rates Table with 3 Reactants**[5 Marks]**
WCH14/01/Hard/Q28

Initial rate data for $A + B + C \rightarrow \text{Products}$:

Exp 1: $[A]=0.10$, $[B]=0.10$, $[C]=0.10$, $\text{Rate}=1.5 \times 10^{-3}$

Exp 2: $[A]=0.20$, $[B]=0.10$, $[C]=0.10$, $\text{Rate}=6.0 \times 10^{-3}$

Exp 3: $[A]=0.10$, $[B]=0.30$, $[C]=0.10$, $\text{Rate}=4.5 \times 10^{-3}$

Exp 4: $[A]=0.10$, $[B]=0.10$, $[C]=0.20$, $\text{Rate}=1.5 \times 10^{-3}$

(a) Determine the order w.r.t A, B, and C with full mathematical justification. [4]

(b) Calculate k including units. [1]

Q29. 29. A* Challenge: Half-Life Determination for 2nd Order Kinetics**[5 Marks]**
WCH14/01/Hard/Q29

For a second-order reaction $2A \rightarrow B$, $\text{rate} = k[A]^2$ where $k = 0.40 \text{ dm}^3 \text{ mol}^{-1} \text{ s}^{-1}$.

(a) Show that the half-life for a second-order reaction is given by $t_{1/2} = 1 / (k[A]_0)$. [3]

(b) Calculate the first half-life when $[A]_0 = 0.50 \text{ mol dm}^{-3}$ and the second half-life as $[A]$ falls from 0.25 to $0.125 \text{ mol dm}^{-3}$. [2]

Q30. 30. A* Challenge: Arrhenius Graph Data Interpretation with Table**[5 Marks]**
WCH14/01/Hard/Q30

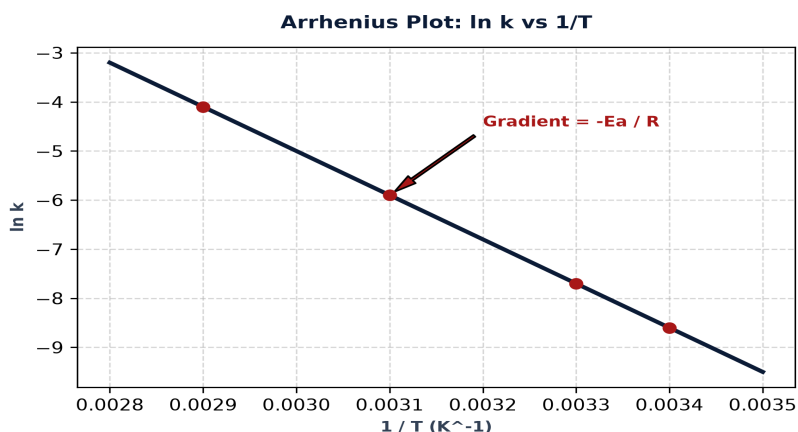
Experimental data for reaction rate constant k at various temperatures T:

$T=290\text{K}$ ($1/T=0.00345$, $\ln k=-8.25$)

$T=310\text{K}$ ($1/T=0.00323$, $\ln k=-5.80$)

$T=330\text{K}$ ($1/T=0.00303$, $\ln k=-3.58$)

$T=350\text{K}$ ($1/T=0.00286$, $\ln k=-1.68$)



(a) Calculate the gradient of the $\ln k$ vs $1/T$ line. [2]

(b) Determine the activation energy, E_a , in kJ mol^{-1} and pre-exponential factor A . [3]

Q31. 31. A* Challenge: Iodine Clock Reaction Kinetics (Persulfate & Iodide)

[5 Marks]
WCH14/01/Hard/Q31

In an iodine clock reaction: $\text{S}_2\text{O}_8^{2-} + 2\text{I}^- \rightarrow 2\text{SO}_4^{2-} + \text{I}_2$

Fixed amount of $\text{Na}_2\text{S}_2\text{O}_3$ and starch indicator added. Time t taken for blue-black colour to appear is recorded.

(a) Explain why $1/t$ can be used as a measure of initial rate of reaction. [2]

(b) State the role of $\text{Na}_2\text{S}_2\text{O}_3$ in the clock reaction. [2]

(c) Why must the amount of $\text{Na}_2\text{S}_2\text{O}_3$ be small relative to reactants? [1]

Q32. 32. A* Challenge: Gas Volume Kinetics Data Analysis

[5 Marks]
WCH14/01/Hard/Q32

Decomposition of benzenediazonium chloride: $\text{C}_6\text{H}_5\text{N}_2\text{Cl(aq)} \rightarrow \text{C}_6\text{H}_5\text{Cl(l)} + \text{N}_2\text{(g)}$

At $t=0\text{s}$ ($V=0\text{cm}^3$), $t=100\text{s}$ ($V=18\text{cm}^3$), $t=200\text{s}$ ($V=32\text{cm}^3$), $t=\text{infinity}$ ($V=58\text{cm}^3$).

(a) Calculate $(V_{\infty} - V_t)$ at $t=0$, 100, and 200 s, which is proportional to reactant concentration. [2]

(b) Prove that the reaction is first order by calculating constant half-life or rate constants. [3]

Q33. 33. A* Challenge: Acid-Catalysed Hydrolysis Rate Law & Solvent Excess

[5 Marks]
WCH14/01/Hard/Q33

The hydrolysis of methyl ethanoate in large excess of water:
 $\text{CH}_3\text{COOCH}_3 + \text{H}_2\text{O} \rightarrow \text{CH}_3\text{COOH} + \text{CH}_3\text{OH}$ (catalysed by H^+).
Overall rate law: $\text{rate} = k' [\text{CH}_3\text{COOCH}_3][\text{H}_2\text{O}][\text{H}^+]$.

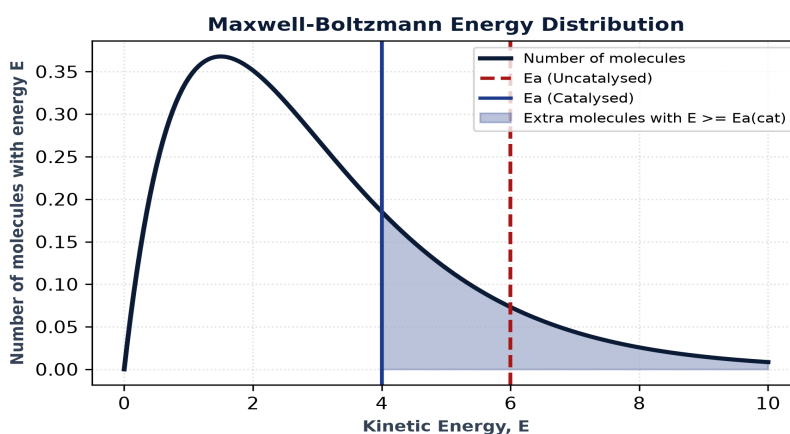
(a) Explain why the reaction behaves as pseudo-first order w.r.t methyl ethanoate. [3]

(b) Relate the observed rate constant k_{obs} to k' , $[\text{H}_2\text{O}]$, and $[\text{H}^+]$. [2]

Q34. 34. A* Challenge: Boltzmann Distribution & Activation Energy Shift

[5 Marks]
WCH14/01/Hard/Q34

A reaction has $E_a = 100 \text{ kJ mol}^{-1}$. Rate doubles when temperature increases from 300 K to 310 K.



(a) Use the Arrhenius equation to calculate the exact ratio of rate constants k_{310} / k_{300} . [3]

- (b) Explain why the fraction of collisions with $E \geq E_a$ increases more significantly for high E_a reactions. [2]

Q35. 35. A* Challenge: Initial Rates with Stoichiometry Considerations

[5 Marks]
WCH14/01/Hard/Q35

For $2A + 3B \rightarrow C + 2D$, the rate of disappearance of A is $4.0 \times 10^{-3} \text{ mol dm}^{-3} \text{ s}^{-1}$.

- (a) Calculate the rate of disappearance of B and the rate of formation of C and D. [3]

- (b) Define reaction rate in terms of $d[C]/dt$. [2]

Q36. 36. A* Challenge: Homogeneous Catalysis Mechanism of $\text{S}_2\text{O}_8^{2-}$ and I^-

[5 Marks]
WCH14/01/Hard/Q36

The reaction $\text{S}_2\text{O}_8^{2-} + 2\text{I}^- \rightarrow 2\text{SO}_4^{2-} + \text{I}_2$ is catalysed by $\text{Fe}^{3+}(\text{aq})$ ions.

- (a) Write two step-wise equations showing how Fe^{3+} acts as a catalyst. [3]

- (b) Explain why catalysis by Fe^{3+} is faster than the uncatalysed reaction. [2]

Q37. 37. A* Challenge: Rate Law Determination from Multi-Step Mechanism with Fast Pre-Equilibrium

[5 Marks]
WCH14/01/Hard/Q37

Mechanism:

Step 1: $A + B \rightleftharpoons AB$ (fast equilibrium, $K_1 = [AB]/([A][B])$)

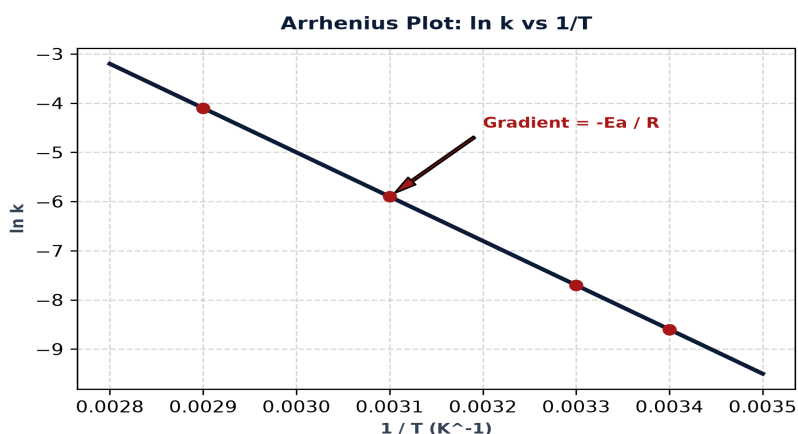
Step 2: $AB + A \rightarrow A_2B$ (slow, k_2)

- (a) Deduce the rate equation. [3]

- (b) What is the order w.r.t A and B? [2]

Q38. 38. A* Challenge: Arrhenius Activation Energy from Arrhenius Equation Ratio**[5 Marks]****WCH14/01/Hard/Q38**

If a reaction rate triples ($k_2/k_1 = 3.0$) when temperature increases from 298 K to 318 K, calculate E_a .



(a) Calculate E_a in kJ mol^{-1} using $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$. [5]

Q39. 39. A* Challenge: Colorimetry Calibration Curve & Kinetics Data Analysis**[5 Marks]****WCH14/01/Hard/Q39**

A colorimeter was calibrated using standard iodine solutions. Absorbance A was proportional to $[I_2]$. At $t=0\text{s}$ ($A=1.00$), $t=200\text{s}$ ($A=0.70$), $t=400\text{s}$ ($A=0.40$), $t=600\text{s}$ ($A=0.10$).

(a) Deduce the order w.r.t I_2 from the constant rate of decrease of absorbance. [3]

(b) Calculate the rate of reaction in absorbance units per second. [2]

Q40. 40. A* Challenge: Quenching Kinetics of Ester Alkaline Hydrolysis**[5 Marks]****WCH14/01/Hard/Q40**

Samples of 25.0 cm³ of a hydrolysing ethyl ethanoate mixture were quenched in ice-cold water and titrated with 0.100 M HCl.

t=0 min: V_{HCl} = 24.0 cm³; t=10 min: V_{HCl} = 16.0 cm³; t=20 min: V_{HCl} = 12.0 cm³; t=40 min: V_{HCl} = 8.0 cm³.

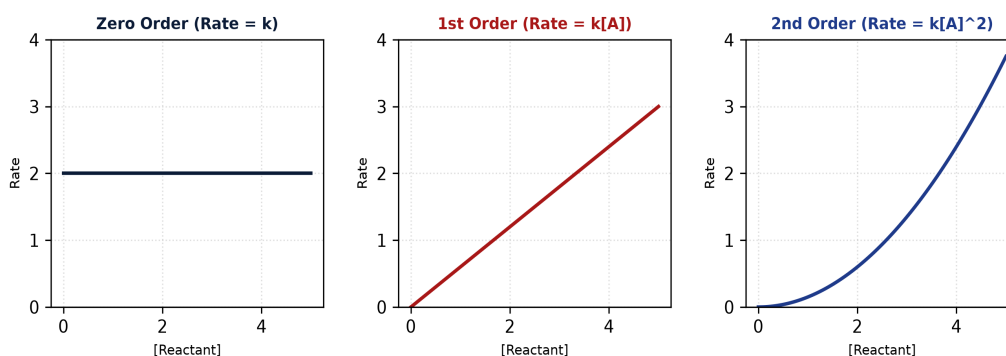
(a) Calculate [OH⁻] at each time. [2]

(b) Show that the reaction is second order overall (1st order in ester, 1st order in OH⁻). [3]

Q41. 41. A* Challenge: Integrated Rate Law Calculation for 1st Order Kinetics

[5 Marks]
WCH14/01/Hard/Q41

The first-order decomposition of N₂O₅ has $k = 3.38 \times 10^{-5} \text{ s}^{-1}$ at 298 K.
Initial concentration [N₂O₅]₀ = 0.500 mol dm⁻³.



(a) Calculate [N₂O₅] remaining after 5.00 hours (18000 s). [3]

(b) Calculate the percentage of N₂O₅ decomposed after 5.00 hours. [2]

Q42. 42. A* Challenge: Temperature Coefficient Q10 Calculation

[5 Marks]
WCH14/01/Hard/Q42

The temperature coefficient Q₁₀ is the factor by which rate increases for a 10 K rise in temperature.
For a reaction with $E_a = 50 \text{ kJ mol}^{-1}$ between 298 K and 308 K.

(a) Calculate $Q_{10} = k_{308} / k_{298}$ using the Arrhenius equation. [4]

(b) Why is Q_{10} approximately equal to 2 for many reactions near room temperature? [1]

**Q43. 43. A* Challenge: Continuous Pressure Measurement
Kinetics of Gas Phase Reaction**

[5 Marks]
WCH14/01/Hard/Q43

Decomposition of azomethane: $\text{CH}_3\text{N}_2\text{CH}_3(\text{g}) \rightarrow \text{C}_2\text{H}_6(\text{g}) + \text{N}_2(\text{g})$

Initial pressure $P_0 = 200$ mmHg. Total pressure P_{total} at time t : $t=0$ (200), $t=100\text{s}$ (250), $t=200\text{s}$ (287.5), $t=\infty$ (400).

(a) Express partial pressure of azomethane P_{azo} in terms of P_0 and P_{total} . [2]

(b) Show that the reaction is first order by half-life analysis of P_{azo} . [3]

**Q44. 44. A* Challenge: Initial Rates Deduction with Fractional
Orders**

[5 Marks]
WCH14/01/Hard/Q44

Reaction: $\text{CHCl}_3(\text{g}) + \text{Cl}_2(\text{g}) \rightarrow \text{CCl}_4(\text{g}) + \text{HCl}(\text{g})$

Exp 1: $[\text{CHCl}_3]=0.10$, $[\text{Cl}_2]=0.10$, $\text{Rate}=1.0 \times 10^{-4}$

Exp 2: $[\text{CHCl}_3]=0.20$, $[\text{Cl}_2]=0.10$, $\text{Rate}=2.0 \times 10^{-4}$

Exp 3: $[\text{CHCl}_3]=0.10$, $[\text{Cl}_2]=0.40$, $\text{Rate}=2.0 \times 10^{-4}$

(a) Deduce the order w.r.t CHCl_3 and Cl_2 (note: Cl_2 order is 0.5). [4]

(b) Write the rate equation. [1]

**Q45. 45. A* Challenge: Multi-Step Mechanism Derivation with
Intermediate Steady State**

[5 Marks]
WCH14/01/Hard/Q45

Ozone decomposition: $2\text{O}_3 \rightarrow 3\text{O}_2$

Step 1: $\text{O}_3 \rightleftharpoons \text{O}_2 + \text{O}$ (fast equilibrium, K_1)

Step 2: $\text{O} + \text{O}_3 \rightarrow 2\text{O}_2$ (slow, k_2)

(a) Deduce the rate equation $\text{rate} = k [\text{O}_3]^2 [\text{O}_2]^{-1}$. [4]

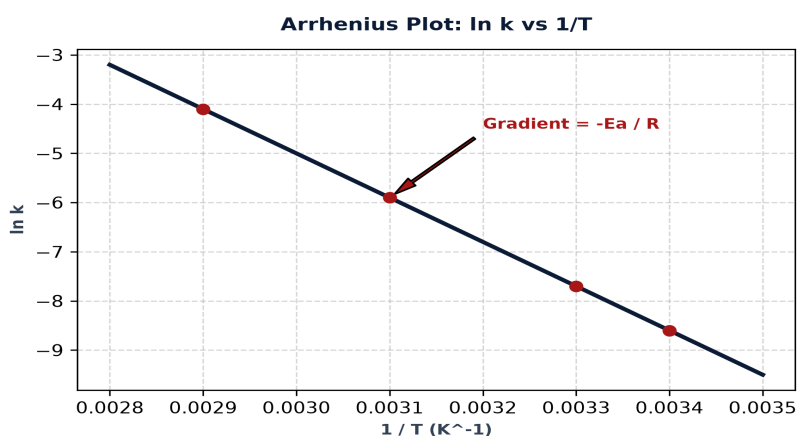
(b) Explain what negative order w.r.t O_2 signifies. [1]

Q46. 46. A* Challenge: Arrhenius Activation Energy Bar Graph & Energy Profile Comparison

[5 Marks]

WCH14/01/Hard/Q46

Compare the activation energy profile of uncatalysed vs catalysed decomposition of H_2O_2 .



(a) Draw an enthalpy profile diagram for exothermic decomposition $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$ showing $E_a(\text{uncat})$ and $E_a(\text{cat})$. [3]

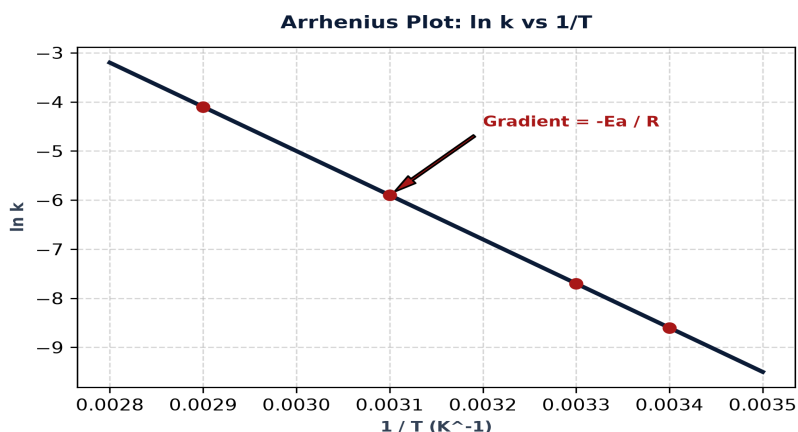
(b) Explain how I^- ions act as a homogeneous catalyst via IO^- intermediate. [2]

Q47. 47. A* Challenge: Arrhenius Line Plot with Experimental Errors

[5 Marks]

WCH14/01/Hard/Q47

An Arrhenius plot for the hydrolysis of ethyl ethanoate has a slope of -5800 K .



(a) Calculate E_a in kJ mol^{-1} . [2]

(b) Discuss how a $\pm 1.0 \text{ K}$ error in temperature measurement affects the calculated E_a at high vs low temperatures. [3]

Q48. 48. A* Challenge: Clock Reaction Order Determination via Variable Initial Concentrations

[5 Marks]

WCH14/01/Hard/Q48

In a persulfate clock reaction, varying $[I^-]$ while keeping $[S_2O_8^{2-}]$ constant gave times t : $[I^-] = 0.05 \text{ M}$ ($t = 160 \text{ s}$), $[I^-] = 0.10 \text{ M}$ ($t = 80 \text{ s}$), $[I^-] = 0.20 \text{ M}$ ($t = 40 \text{ s}$).

(a) Calculate initial rates ($1/t$) and deduce the order w.r.t I^- . [3]

(b) Predict the time t if $[I^-] = 0.15 \text{ M}$. [2]

Q49. 49. A* Challenge: Complex Reaction Mechanism with Enzyme Michaelis-Menten Kinetics

[5 Marks]

WCH14/01/Hard/Q49

Enzyme kinetics: $E + S \rightleftharpoons ES$ (fast equilibrium) $\rightarrow E + P$ (slow, k_2).

(a) Explain why at low $[S]$, reaction is 1st order w.r.t S , but at high $[S]$, reaction becomes 0 order w.r.t S . [3]

(b) Define $V_{\max}$ and K_m . [2]

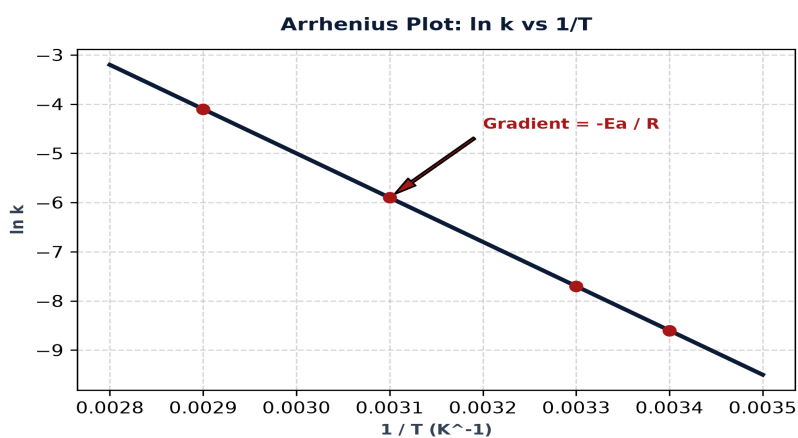
Q50. 50. A* Challenge: Complete Rate Law, Mechanism & Arrhenius Synthesis

[6 Marks]
WCH14/01/Hard/Q50

The gas-phase reaction $2\text{NO}_2 + \text{F}_2 \rightarrow 2\text{NO}_2\text{F}$ was investigated.

Exp data: 1st order in NO_2 , 1st order in F_2 .

At 300 K, $k = 0.80 \text{ dm}^3 \text{ mol}^{-1} \text{ s}^{-1}$; at 350 K, $k = 12.0 \text{ dm}^3 \text{ mol}^{-1} \text{ s}^{-1}$.



(a) Write the rate equation. [1]

(b) Propose a valid two-step mechanism. [2]

(c) Calculate the activation energy E_a in kJ mol^{-1} . [3]

SECTION C: WORKED MARK SCHEME (Q1–Q50) — 11A

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. Continuous Monitoring of Reaction Rate 1. (a) Measure volume of H ₂ gas using a gas syringe over time (1). H ₂ is the only gas produced while reactants are solid/liquid (1). 1. (b) Draw a tangent to the curve at t = 0 seconds (1). Calculate gradient of tangent = Change in volume / Change in time (1).	[4]
Q2	Q2: 2. Colorimetric Measurement of Reaction Rate 2. (a) Iodine is brown/yellow while all other species are colourless (1). Absorbance decreases proportionally as iodine is consumed (1). 2. (b) Allows continuous non-destructive monitoring without disturbing the reaction mixture (1).	[3]
Q3	Q3: 3. Quenching and Titrimetry 3. (a) Quenching halts or drastically slows the reaction (1) so OH ⁻ concentration does not continue to change during titration (1). 3. (b) Pipette sample into ice-cold water (1) or add a known excess of ice-cold acid (1).	[4]
Q4	Q4: 4. Electrical Conductivity Method 4. (a) Number of mobile ions increases from 1 mole to 3 moles of ions (2 H ⁺ and 1 Br ⁻) (1); H ⁺ has exceptionally high ionic mobility (1). 4. (b) Plot conductivity vs time and find the gradient at t=0 (1).	[3]
Q5	Q5: 5. Rate Equation and Rate Constant Units 5. (a) Order w.r.t A = 1, w.r.t B = 2 (1); Overall order = 1 + 2 = 3 (1). 5. (b) units of k = (mol dm ⁻³ s ⁻¹) / ((mol dm ⁻³)(mol dm ⁻³) ²) = dm ⁶ mol ⁻² s ⁻¹ (1).	[3]
Q6	Q6: 6. Zero Order Kinetics Definition 6. (a) X is not involved in the rate-determining step (1). 6. (b) Straight line with constant negative gradient (1).	[2]
Q7	Q7: 7. First Order Half-Life Relationship 7. (a) Time taken for the concentration of a reactant to fall to half its initial value (1). 7. (b) The half-life is constant (1) regardless of the starting concentration (1).	[3]
Q8	Q8: 8. Calculation of Rate Constant from Half-Life 8. (a) $k = 0.69315 / 450 = 1.54 \times 10^{-3} \text{ s}^{-1}$ (2). 8. (b) rate = $(1.54 \times 10^{-3})(0.080) = 1.23 \times 10^{-4} \text{ mol dm}^{-3} \text{ s}^{-1}$ (1).	[3]
Q9	Q9: 9. Initial Rates Method: Deduction of Orders 9. (a) Exp 1→2: [NO] x2, [O ₂] constant → Rate x4 (2 ²) ⇒ 2nd order w.r.t NO (2). Exp 1→3: [O ₂] x3, [NO] constant → Rate x3 (3 ¹) ⇒ 1st order w.r.t O ₂ (1). 9. (b) rate = $k[\text{NO}]^2[\text{O}_2]$ (1).	[4]
Q10	Q10: 10. Calculation of Rate Constant k from Experimental Table 10. (a) $k = (1.4 \times 10^{-4}) / ((0.010)^2(0.010)) = 14000 \text{ dm}^6 \text{ mol}^{-2} \text{ s}^{-1}$ (2). 10. (b) dm ⁶ mol ⁻² s ⁻¹ (1).	[3]
Q11	Q11: 11. Rate-Determining Step Deduction from Rate Equation 11. (a) CO is not involved in the rate-determining step (RDS) or acts in a fast step after RDS (2). 11. (b) Step 1: NO ₂ + NO ₂ → NO ₃ + NO (slow) (1). Step 2: NO ₃ + CO → NO ₂ + CO ₂ (fast) (1).	[4]
Q12	Q12: 12. Catalysis & Rate Equation Inclusion 12. (a) H ⁺ is a homogeneous catalyst involved in the rate-determining step (2). 12. (b) Zero order w.r.t I ₂ (1).	[3]
Q13	Q13: 13. Initial Rates Table with Fractional / Higher Orders 13. (a) Exp 1→2: [A] x2, Rate x4 → 2nd order w.r.t A (2). Exp 1→3: [B] x2, Rate unchanged → 0 order w.r.t B (1). 13. (b) $k = (2.0 \times 10^{-3}) / (0.20)^2 = 0.050 \text{ dm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ (1).	[4]

Q14	Q14: 14. Multi-Step Mechanism Validation 14. (a) From Step 1 equilibrium: $[N_2O_2] = K[NO]^2$ (1). Step 2 is RDS: $rate = k_2[N_2O_2][H_2]$ (1). Substitute $[N_2O_2]$: $rate = k[NO]^2[H_2]$ (1). 14. (b) Intermediates: N_2O_2 and N_2O (1).	[4]
Q15	Q15: 15. Rate-Concentration Graph Summary 15. (a) Zero order: Horizontal line (rate independent of $[A]$) (1). 1st order: Straight line through origin (rate proportional to $[A]$) (1). 2nd order: Parabolic curve upwards (rate proportional to $[A]^2$) (1).	[3]
Q16	Q16: 16. Hydrolysis of Tertiary Halogenoalkanes Mechanism 16. (a) Hydrolysis proceeds via S_N1 mechanism where heterolytic fission of C-Br bond to form carbocation is slow (1). Nucleophilic attack by OH^- occurs in a fast step (1). 16. (b) $(CH_3)_3CBr \rightarrow (CH_3)_3C^+ + Br^-$ (slow) (2).	[4]
Q17	Q17: 17. Hydrolysis of Primary Halogenoalkanes Mechanism 17. (a) Both CH_3CH_2Br and OH^- are involved in the single rate-determining step (2). 17. (b) Five-coordinate transition state (1) with partial C...OH bond formation and partial C...Br bond cleavage (1).	[4]
Q18	Q18: 18. Effect of Temperature on Rate Constant k 18. (a) Significantly larger fraction of molecules have energy $E \geq E_a$ (Maxwell-Boltzmann distribution) (2). 18. (b) Rate constant k increases exponentially with temperature (1).	[3]
Q19	Q19: 19. Arrhenius Equation Formulation 19. (a) A = pre-exponential frequency factor, E_a = activation energy ($J\ mol^{-1}$), R = gas constant ($8.31\ J\ K^{-1}\ mol^{-1}$), T = temperature in Kelvin (2). 19. (b) $\ln k = -E_a / (RT) + \ln A$ (1).	[3]
Q20	Q20: 20. Arrhenius Plot Graphical Analysis: N_2O_5 Decomposition 20. (a) Gradient = $-E_a / R = -11000\ K$ (1). $E_a = 11000 \times 8.31 = 91410\ J\ mol^{-1}$ (1) = $+91.4\ kJ\ mol^{-1}$ (1). 20. (b) $\ln A = 29.4 \Rightarrow A = e^{29.4} = 5.86 \times 10^{12}\ s^{-1}$ (2).	[5]
Q21	Q21: 21. Heterogeneous Catalysis Surface Adsorption 21. (a) Adsorption of N_2 and H_2 onto active sites on iron surface (1); weakening of $N=N$ and $H-H$ bonds (1); reaction forms NH_3 followed by desorption of NH_3 product (1).	[3]
Q22	Q22: 22. Homogeneous Catalysis Mechanism: Fe^{2+} and $S_2O_8^{2-}$ 22. (a) High activation energy due to electrostatic repulsion between two negatively charged ions (1). 22. (b) Step 1: $S_2O_8^{2-} + 2Fe^{2+} \rightarrow 2SO_4^{2-} + 2Fe^{3+}$ (2). Step 2: $2Fe^{3+} + 2I^- \rightarrow 2Fe^{2+} + I_2$ (1).	[4]
Q23	Q23: 23. Maxwell-Boltzmann Temperature Shift Analysis 23. (a) Total number of molecules remains constant (area under curve equal) (1); average kinetic energy increases, spreading the distribution (1). 23. (b) Area to the right of E_a line is significantly greater at T_2 , resulting in a much higher frequency of successful collisions (2).	[4]
Q24	Q24: 24. Autocatalysis Mechanism: MnO_4^- and $C_2O_4^{2-}$ 24. (a) Autocatalysis is when one of the products of the reaction acts as a catalyst for the reaction (1). 24. (b) Initially slow due to repulsion between MnO_4^- and $C_2O_4^{2-}$ (1); rate speeds up rapidly as Mn^{2+} catalyst is generated (1); rate slows down near completion as reactants are exhausted (1).	[4]
Q25	Q25: 25. Maxwell-Boltzmann Catalysed Activation Energy Barrier 25. (a) Catalyst provides an alternative pathway with lower activation energy $E_a(cat)$ (1). Position of E_a line shifts left (1). A larger fraction of molecules have $E \geq E_a(cat)$, increasing successful collision frequency (1).	[3]
Q26	Q26: 26. A* Challenge: Two-Point Arrhenius Activation Energy Calculation 26. (a) $\ln(1.25 \times 10^{-4} / 2.50 \times 10^{-4}) = \ln(50) = 3.912$ (1). $(1/300 - 1/340) = 3.922 \times 10^{-4}\ K^{-1}$ (1). $E_a/R = 3.912 / 3.922 \times 10^{-4} = 9975\ K$ (1). $E_a = 9975 \times 8.31 = 82.9\ kJ\ mol^{-1}$ (1). 26. (b) $A = k / e^{(-E_a/RT)} = (2.50 \times 10^{-4}) / e^{(-8290/(8.31 \times 300))} = 6.64 \times 10^{10}\ s^{-1}$ (1).	[5]

Q27	Q27: 27. A* Challenge: Deduction of Complex Rate Equation from Mechanism 27. (a) Step 2 is RDS: rate = $k_2[A_2][B]$ (1). From Step 1 equilibrium: $[A_2] = K_1[A]^2$ (1). Substitute: rate = $k_2 K_1 [A]^2 [B] = k[A]^2[B]$ (1). 27. (b) Overall order = 3 (1). Units of $k = \text{dm}^6 \text{mol}^{-2} \text{s}^{-1}$ (1).	[5]
Q28	Q28: 28. A* Challenge: Initial Rates Table with 3 Reactants 28. (a) Exp 1→2: $[A]$ x2, Rate x4 ⇒ 2nd order w.r.t A (1). Exp 1→3: $[B]$ x3, Rate x3 ⇒ 1st order w.r.t B (1). Exp 1→4: $[C]$ x2, Rate unchanged ⇒ 0 order w.r.t C (1). Rate equation = $k[A]^2[B]$ (1). 28. (b) $k = (1.5 \times 10^{-3}) / ((0.10)^2(0.10)) = 1.50 \text{ dm}^6 \text{mol}^{-2} \text{s}^{-1}$ (1).	[5]
Q29	Q29: 29. A* Challenge: Half-Life Determination for 2nd Order Kinetics 29. (a) Integrated 2nd order rate law: $1/[A] - 1/[A]_0 = k t$ (1). At $t = t_{1/2}$, $[A] = [A]_0/2 \Rightarrow 2/[A]_0 - 1/[A]_0 = k t_{1/2}$ (1) ⇒ $t_{1/2} = 1 / (k[A]_0)$ (1). 29. (b) 1st $t_{1/2} = 1 / (0.40 \times 0.50) = 5.0 \text{ s}$ (1). 2nd $t_{1/2} = 1 / (0.40 \times 0.25) = 10.0 \text{ s}$ (1). Half-life doubles as concentration halves.	[5]
Q30	Q30: 30. A* Challenge: Arrhenius Graph Data Interpretation with Table 30. (a) Gradient = $(-1.68 - (-8.25)) / (0.00286 - 0.00345) = 6.57 / (-0.00059) = -11135 \text{ K}$ (2). 30. (b) $E_a = 11135 \times 8.31 = 92530 \text{ J mol}^{-1} = +92.5 \text{ kJ mol}^{-1}$ (2). $\ln A = -1.68 + (11135 \times 0.00286) = 30.16 \Rightarrow A = e^{30.16} = 1.25 \times 10^{13} \text{ s}^{-1}$ (1).	[5]
Q31	Q31: 31. A* Challenge: Iodine Clock Reaction Kinetics (Persulfate & Iodide) 31. (a) $1/t$ is proportional to rate because the amount of I_2 produced to reach the blue-black endpoint is constant (2). 31. (b) Thiosulfate reacts immediately with I_2 as it forms: $2S_2O_3^{2-} + I_2 \rightarrow S_4O_6^{2-} + 2I^-$ (1), delaying the appearance of starch-iodine blue-black colour (1). 31. (c) Ensures initial rate is measured before significant reactant depletion (1).	[5]
Q32	Q32: 32. A* Challenge: Gas Volume Kinetics Data Analysis 32. (a) At $t=0$: $58 - 0 = 58 \text{ cm}^3$ (1). At $t=100$: $58 - 18 = 40 \text{ cm}^3$. At $t=200$: $58 - 32 = 26 \text{ cm}^3$ (1). 32. (b) $k_1 = (1/100) \ln(58/40) = 0.00372 \text{ s}^{-1}$ (1). $k_2 = (1/200) \ln(58/26) = 0.00401 \text{ s}^{-1}$ (1). k is approximately constant ⇒ 1st order kinetics (1).	[5]
Q33	Q33: 33. A* Challenge: Acid-Catalysed Hydrolysis Rate Law & Solvent Excess 33. (a) $[H_2O]$ is in large excess so its concentration remains effectively constant (1). $[H^+]$ is a catalyst so its concentration is constant (1). Therefore rate = $k_{\text{obs}} [CH_3COOCH_3]$ where $k_{\text{obs}} = k' [H_2O][H^+]$ (1). 33. (b) $k_{\text{obs}} = k' [H_2O][H^+]$ (2).	[5]
Q34	Q34: 34. A* Challenge: Boltzmann Distribution & Activation Energy Shift 34. (a) $\ln(k_{310}/k_{300}) = (100000/8.31) * (1/300 - 1/310) = 12033.7 * (0.001075) = 1.294$ (2). $k_{310}/k_{300} = e^{1.294} = 3.65$ (1). 34. (b) High E_a sits far out on the tail of the Boltzmann curve where the relative percentage increase in area under curve upon heating is largest (2).	[5]
Q35	Q35: 35. A* Challenge: Initial Rates with Stoichiometry Considerations 35. (a) $-d[B]/dt = 1.5 \times (-d[A]/dt) = 6.0 \times 10^{-3} \text{ mol dm}^{-3} \text{s}^{-1}$ (1). $d[C]/dt = 0.5 \times 4.0 \times 10^{-3} = 2.0 \times 10^{-3} \text{ mol dm}^{-3} \text{s}^{-1}$ (1). $d[D]/dt = 4.0 \times 10^{-3} \text{ mol dm}^{-3} \text{s}^{-1}$ (1). 35. (b) Overall rate = $d[C]/dt = -1/2 d[A]/dt = -1/3 d[B]/dt = 1/2 d[D]/dt$ (2).	[5]
Q36	Q36: 36. A* Challenge: Homogeneous Catalysis Mechanism of $S_2O_8^{2-}$ and I^- 36. (a) Step 1: $2Fe^{3+} + 2I^- \rightarrow 2Fe^{2+} + I_2$ (2). Step 2: $2Fe^{2+} + S_2O_8^{2-} \rightarrow 2Fe^{3+} + 2SO_4^{2-}$ (1). 36. (b) Both steps involve reaction between oppositely charged ions (Fe^{3+} and I^- , Fe^{2+} and $S_2O_8^{2-}$) which attract, whereas uncatalysed reaction is between two negatively charged ions ($S_2O_8^{2-}$ and I^-) which repel (2).	[5]
Q37	Q37: 37. A* Challenge: Rate Law Determination from Multi-Step Mechanism with Fast Pre-Equilibrium 37. (a) RDS is Step 2: rate = $k_2[AB][A]$ (1). From Step 1 equilibrium: $[AB] = K_1[A][B]$ (1). Substitute: rate = $k_2 K_1 [A]^2 [B] = k[A]^2[B]$ (1). 37. (b) Order w.r.t A = 2, w.r.t B = 1 (2).	[5]
Q38	Q38: 38. A* Challenge: Arrhenius Activation Energy from Arrhenius Equation Ratio 38. (a) $\ln(3.0) = 1.0986$ (1). $1/T_1 - 1/T_2 = (1/298 - 1/318) = 3.3556 \times 10^{-4} \text{ K}^{-1}$ (1). $E_a/R = 1.0986 / 3.3556 \times 10^{-4} = 3273.9 \text{ K}$ (2). $E_a = 3273.9 \times 8.31 = 27206 \text{ J mol}^{-1} = +27.2 \text{ kJ mol}^{-1}$ (1).	[5]

Q39	Q39: 39. A* Challenge: Colorimetry Calibration Curve & Kinetics Data Analysis 39. (a) Absorbance decreases linearly over equal time intervals ($\Delta A = 0.30$ every 200 s) (2). Constant rate independent of concentration $\Rightarrow$ Zero order w.r.t I ₂ (1). 39. (b) Rate = $0.30 / 200 = 1.50 \times 10^{-3}$ a.u. s ⁻¹ (2).	[5]
Q40	Q40: 40. A* Challenge: Quenching Kinetics of Ester Alkaline Hydrolysis 40. (a) [OH ⁻] is proportional to V _{HCl} : t=0: 0.096 M; t=10: 0.064 M; t=20: 0.048 M; t=40: 0.032 M (2). 40. (b) Plot 1/[OH ⁻] vs time: t=0 (10.4), t=10 (15.6), t=20 (20.8), t=40 (31.3). Straight line plot confirms 2nd order kinetics (3).	[5]
Q41	Q41: 41. A* Challenge: Integrated Rate Law Calculation for 1st Order Kinetics 41. (a) $\ln([N_2O_5]_t / 0.500) = -k t = -(3.38 \times 10^{-5})(18000) = -0.6084$ (2). $[N_2O_5]_t = 0.500 \times e^{-0.6084} = 0.272$ mol dm ⁻³ (1). 41. (b) Decomposed = $0.500 - 0.272 = 0.228$ M. Percentage = $(0.228 / 0.500) \times 100\% = 45.6\%$ (2).	[5]
Q42	Q42: 42. A* Challenge: Temperature Coefficient Q ₁₀ Calculation 42. (a) $\ln(k_{308}/k_{298}) = (50000 / 8.31) \times (1/298 - 1/308) = 6016.8 \times (1.0888 \times 10^{-4}) = 0.6551$ (3). $Q_{10} = e^{0.6551} = 1.93$ (1). 42. (b) For reactions with $E_a \sim 50\text{--}60$ kJ mol ⁻¹ , a 10 K rise doubles the fraction of molecules with $E \geq E_a$ (1).	[5]
Q43	Q43: 43. A* Challenge: Continuous Pressure Measurement Kinetics of Gas Phase Reaction 43. (a) $CH_3N_2CH_3 \rightarrow C_2H_6 + N_2$. At time t: $P_{\text{azo}} = P_0 - x$, $P_{\text{total}} = P_0 + x \Rightarrow x = P_{\text{total}} - P_0 \Rightarrow P_{\text{azo}} = 2P_0 - P_{\text{total}}$ (2). 43. (b) At t=0: $P_{\text{azo}} = 200$. At t=100: $P_{\text{azo}} = 400 - 250 = 150$. At t=200: $P_{\text{azo}} = 400 - 287.5 = 112.5$. Constant $t_{1/2} = 240$ s $\Rightarrow$ 1st order (3).	[5]
Q44	Q44: 44. A* Challenge: Initial Rates Deduction with Fractional Orders 44. (a) Exp 1 $\rightarrow$ 2: $[CHCl_3] \times 2$, Rate $\times 2 \Rightarrow$ 1st order w.r.t $CHCl_3$ (2). Exp 1 $\rightarrow$ 3: $[Cl_2] \times 4$, Rate $\times 2$ ($4^{0.5}$) $\Rightarrow$ 0.5 order w.r.t Cl_2 (2). 44. (b) rate = $k[CHCl_3][Cl_2]^{0.5}$ (1).	[5]
Q45	Q45: 45. A* Challenge: Multi-Step Mechanism Derivation with Intermediate Steady State 45. (a) Step 2 is RDS: rate = $k_2[O][O_3]$ (1). From Step 1 equilibrium: $[O] = K_1[O_3] / [O_2]$ (2). Substitute: rate = $k_2 K_1 [O_3]^2 / [O_2] = k [O_3]^2 [O_2]^{-1}$ (1). 45. (b) O ₂ acts as an inhibitor by driving the reverse of Step 1 (1).	[5]
Q46	Q46: 46. A* Challenge: Arrhenius Activation Energy Bar Graph & Energy Profile Comparison 46. (a) Reactants (2H ₂ O ₂) higher energy than products (2H ₂ O + O ₂) (1); uncatalysed single high hump $E_a(\text{uncat}) = 75$ kJ mol ⁻¹ (1); catalysed two lower humps $E_a(\text{cat}) = 56$ kJ mol ⁻¹ (1). 46. (b) Step 1: $H_2O_2 + I^- \rightarrow H_2O + IO^-$ (slow) (1). Step 2: $H_2O_2 + IO^- \rightarrow H_2O + O_2 + I^-$ (fast) (1).	[5]
Q47	Q47: 47. A* Challenge: Arrhenius Line Plot with Experimental Errors 47. (a) $E_a = 5800 \times 8.31 = 48200$ J mol ⁻¹ = $+48.2$ kJ mol ⁻¹ (2). 47. (b) Since 1/T is non-linear, a 1 K error at low temperature (e.g. 280 K) produces a larger absolute error in 1/T than at high temperature (e.g. 350 K) (3).	[5]
Q48	Q48: 48. A* Challenge: Clock Reaction Order Determination via Variable Initial Concentrations 48. (a) $1/t = 0.00625, 0.0125, 0.0250$. Doubling $[I^-]$ doubles rate ($1/t$) $\Rightarrow$ 1st order w.r.t I ⁻ (3). 48. (b) Rate is proportional to $[I^-]$: $t = 80 \times (0.10 / 0.15) = 53.3$ s (2).	[5]
Q49	Q49: 49. A* Challenge: Complex Reaction Mechanism with Enzyme Michaelis-Menten Kinetics 49. (a) At low $[S]$, active sites are available so rate = $k_2 [ES] [S]$ (1st order) (1.5). At high $[S]$, all enzyme active sites are saturated so $[ES] = [E]_{\text{total}}$ (constant), making rate independent of $[S]$ (0 order) (1.5). 49. (b) V_{max} = maximum rate at saturation; K_m = substrate concentration at half V_{max} (2).	[5]

Q50	Q50: 50. A* Challenge: Complete Rate Law, Mechanism & Arrhenius Synthesis 50. (a) $\text{rate} = k[\text{NO}_2][\text{F}_2]$ (1). 50. (b) Step 1: $\text{NO}_2 + \text{F}_2 \rightarrow \text{NO}_2\text{F} + \text{F}$ (slow, RDS) (1). Step 2: $\text{NO}_2 + \text{F} \rightarrow \text{NO}_2\text{F}$ (fast) (1). 50. (c) $\ln(12.0 / 0.80) = \ln(15) = 2.708$ (1). $(1/300 - 1/350) = 4.762 \times 10^{-4} \text{ K}^{-1}$ (1). $E_a = (2.708 / 4.762 \times 10^{-4}) \times 8.31 = 47.3 \text{ kJ mol}^{-1}$ (1).	[6]
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SECTION D: EMPIRICAL EXAMINER FAQs & COMMON TRAPS (10 FAQs) — 11A**FAQ #1: Units of Rate Constant k****Category: Calculation Trap****COMMON EXAMINER TRAP / MISTAKE:**

Using $\text{mol dm}^{-3} \text{s}^{-1}$ for k regardless of overall order.

MODEL ANSWER / EXAMINER EXPECTATION:

Substitute units into rate equation: units of $k = (\text{mol dm}^{-3} \text{s}^{-1}) / (\text{mol dm}^{-3})^n$. For 1st order: s^{-1} ; 2nd order: $\text{dm}^3 \text{mol}^{-1} \text{s}^{-1}$; 3rd order: $\text{dm}^6 \text{mol}^{-2} \text{s}^{-1}$.

FAQ #2: Arrhenius Temperature in Kelvin**Category: Units Trap****COMMON EXAMINER TRAP / MISTAKE:**

Inserting temperature in Celsius into the Arrhenius equation.

MODEL ANSWER / EXAMINER EXPECTATION:

Always convert temperature to Kelvin ($K = ^\circ\text{C} + 273.15$). Using Celsius leads to negative or nonsense activation energy values.

FAQ #3: Activation Energy Units (kJ vs J)**Category: Conversion Trap****COMMON EXAMINER TRAP / MISTAKE:**

Forgetting to multiply/divide by 1000 when $R = 8.31 \text{ J K}^{-1} \text{mol}^{-1}$ is used.

MODEL ANSWER / EXAMINER EXPECTATION:

R is given in Joules ($\text{J K}^{-1} \text{mol}^{-1}$). The gradient $-E_a/R$ yields E_a in J mol^{-1} . Divide by 1000 to express final E_a in kJ mol^{-1} .

FAQ #4: Zero Order Reactants in Rate Equation**Category: Mechanism Trap****COMMON EXAMINER TRAP / MISTAKE:**

Including zero-order reactants in the rate equation.

MODEL ANSWER / EXAMINER EXPECTATION:

Zero-order reactants have rate $[X]^0 = 1$. Omit them from the rate equation completely, but note they participate after the RDS.

FAQ #5: Rate-Determining Step (RDS) Identification**Category: Theory Trap****COMMON EXAMINER TRAP / MISTAKE:**

Including species formed in fast steps AFTER the RDS in the rate equation.

MODEL ANSWER / EXAMINER EXPECTATION:

The rate equation only contains species present in or before the rate-determining step. Products of fast subsequent steps do not appear.

FAQ #6: Constant Half-Life Significance**Category: Graph Interpretation****COMMON EXAMINER TRAP / MISTAKE:**

Assuming constant half-life applies to zero or second order reactions.

MODEL ANSWER / EXAMINER EXPECTATION:

ONLY first-order reactions have a constant half-life independent of initial concentration ($t_{1/2} = \ln 2 / k$).

FAQ #7: Initial Rate Method Gradient Tangent**Category: Practical Skills****COMMON EXAMINER TRAP / MISTAKE:**

Drawing tangents at non-zero times when calculating initial rate.

MODEL ANSWER / EXAMINER EXPECTATION:

Initial rate MUST be determined by drawing a tangent at $t = 0$ seconds (when zero product has accumulated).

FAQ #8: Quenching Reaction Samples**Category: Practical Technique****COMMON EXAMINER TRAP / MISTAKE:**

Titration samples directly without quenching first.

MODEL ANSWER / EXAMINER EXPECTATION:

Quenching (ice-cold water/acid) halts or drastically slows the reaction so concentration does not change during titration.

FAQ #9: Catalysts in Rate Equations**Category: Homogeneous Catalysts****COMMON EXAMINER TRAP / MISTAKE:**

Excluding homogeneous catalysts from the rate equation.

MODEL ANSWER / EXAMINER EXPECTATION:

Homogeneous catalysts ARE involved in the rate-determining step and DO appear in the rate equation (e.g. H^+ in acid-catalysed esterification).

FAQ #10: Gradient of Arrhenius Plot**Category: Graph Analysis****COMMON EXAMINER TRAP / MISTAKE:**

Setting gradient equal to $-E_a$ instead of $-E_a/R$.

MODEL ANSWER / EXAMINER EXPECTATION:

On a plot of $\ln k$ against $1/T$, gradient = $-E_a / R$. Therefore $E_a = -\text{gradient} \times 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$.